BIO INSPIRED ALGORITHMS · 23DS7BSBIO

Cover Slide/Main Slide

Plant Disease Detection using Transfer-Learning CNN Optimized with Differential Evolution

INTELLIGENCE, NATURALLY.

Under the guidance of

Manoj Kumar S, Assistant Professor

Academic Year 2026–2027

TEAM 10

Aryanraj S Singh
1BM23CD008

Richik Chaudhuri
1BM23CD050

Suraj B
1BM23CD061

Sushant Deo
1BM23CD062

FINDING THE BEST SETTINGS

Introduction to Optimization Problems

Optimization = finding the settings that give the best result.

01

Settings

Our model has 7 training settings, such as learning speed and how much of the network to retrain.

02

Score

Each set of settings is tested by training a model and scoring it on images it has not seen.

03

Search

Try settings smartly, with a fixed number of attempts, and keep the best one.

LEARNING FROM NATURE

What are Nature-Inspired Algorithms?

Search methods that borrow ideas from nature, such as evolution or swarms.

01

A population

Work with a group of candidate solutions, not just one guess.

02

Variation

Mix and change candidates to create new ones, like offspring.

03

Selection

Keep whichever candidate scores better, like survival of the fittest.

A MAP OF THE FIELD

Classification of Nature-Inspired Algorithms

Our choice, Differential Evolution (DE), belongs to the evolutionary family.

01

Evolutionary

Genetic Algorithm

Differential Evolution (ours)

Evolution Strategies

02

Swarm

Particle Swarm

Ant Colony

Artificial Bee Colony

03

Other families

Simulated Annealing

Biogeography-based

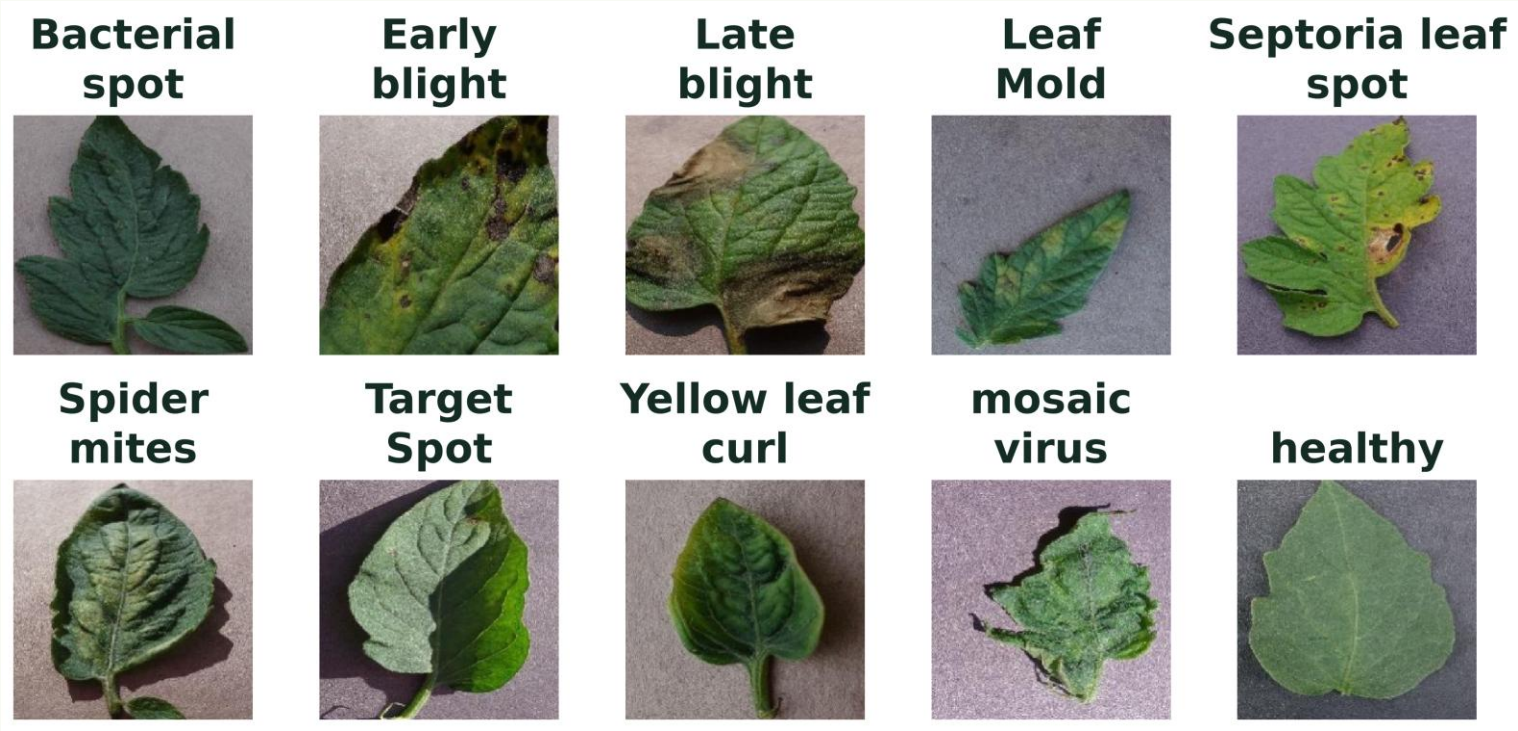
Teaching–Learning

TOMATO LEAF HEALTH

Problem Statement

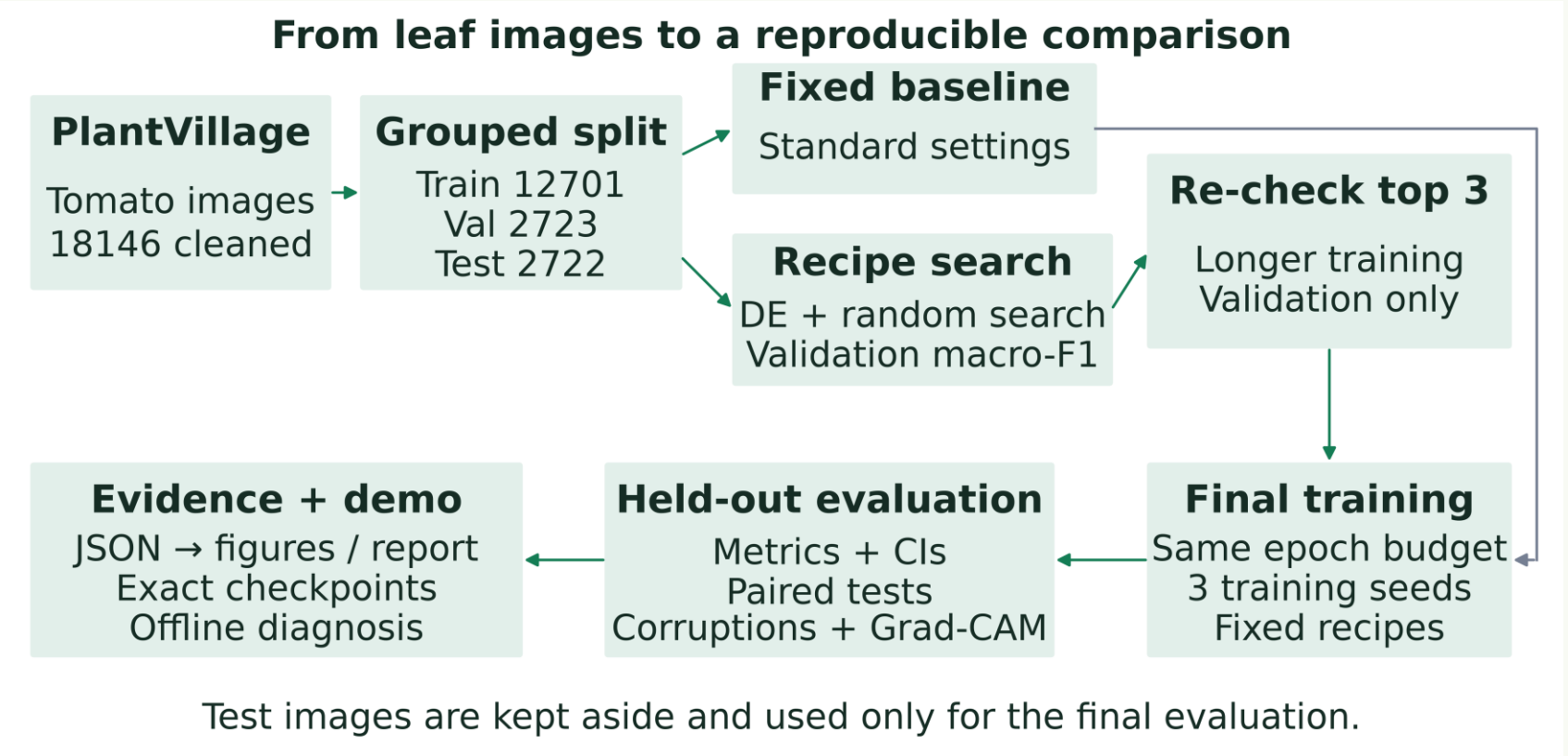
Goal: use DE to pick the training settings that best detect tomato leaf diseases.

18,146 leaf images · 10 classes (9 diseases + healthy) · 70% train / 15% validation / 15% test



THE FULL SYSTEM

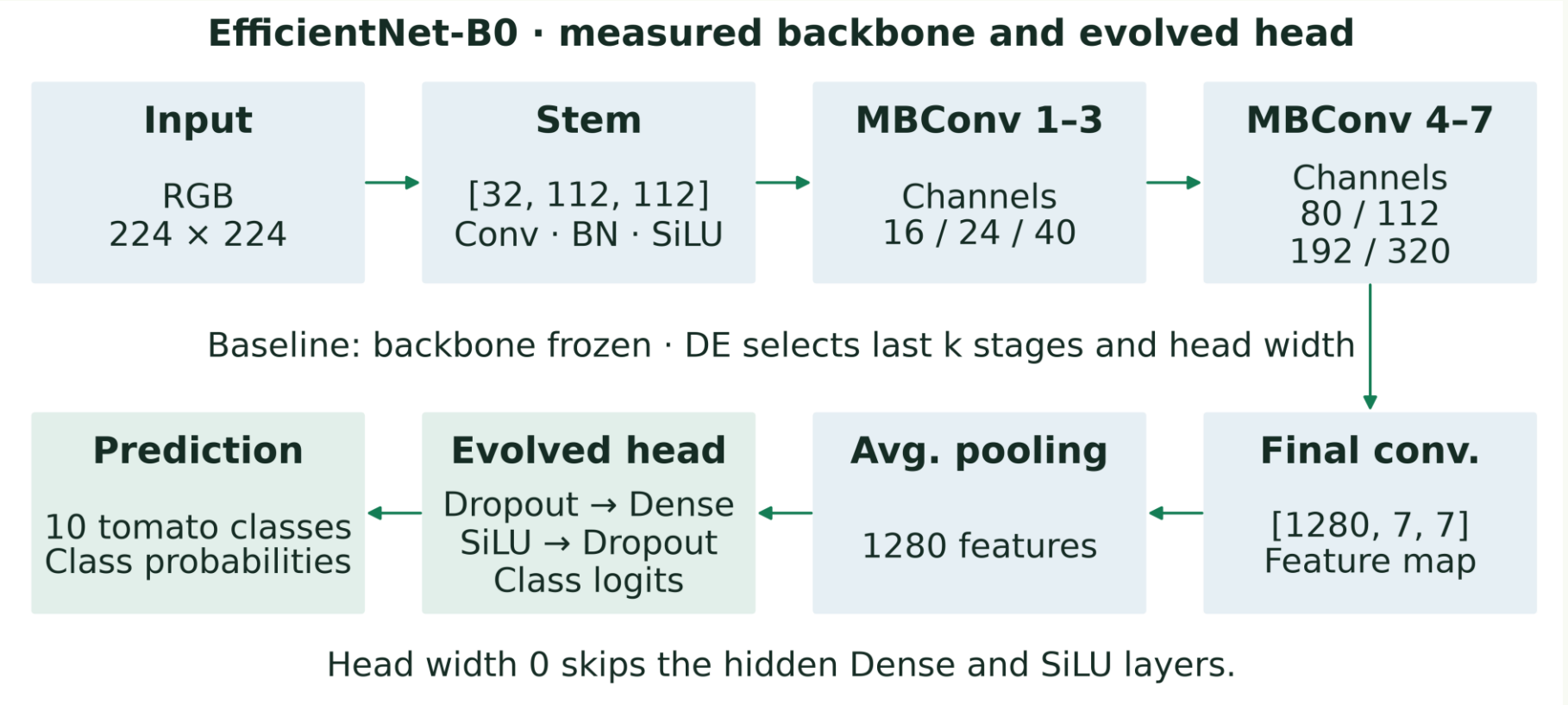
Implementation and Results (1/10)



Test images are kept aside and used only once, for the final score.

OUR MODEL AND THE BASELINE

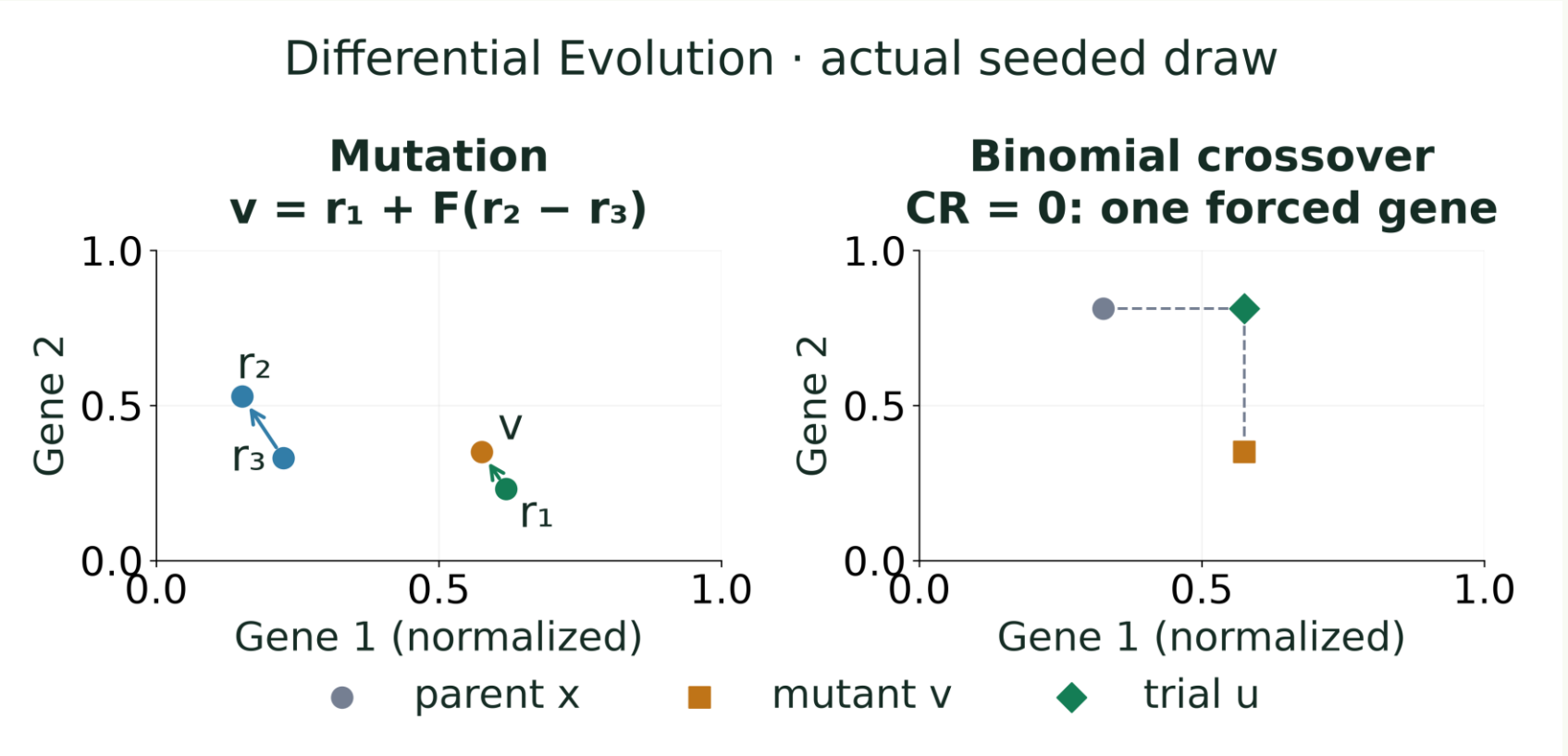
Implementation and Results (2/10)



EfficientNet-B0: a compact, pretrained image network. Baseline = standard recipe fixed in advance.

HOW DE CREATES BETTER CANDIDATES

Implementation and Results (3/10)



Mutate: combine three candidates · Crossover: blend with the parent · Select: keep the better one

THE 7 SETTINGS DE TUNES

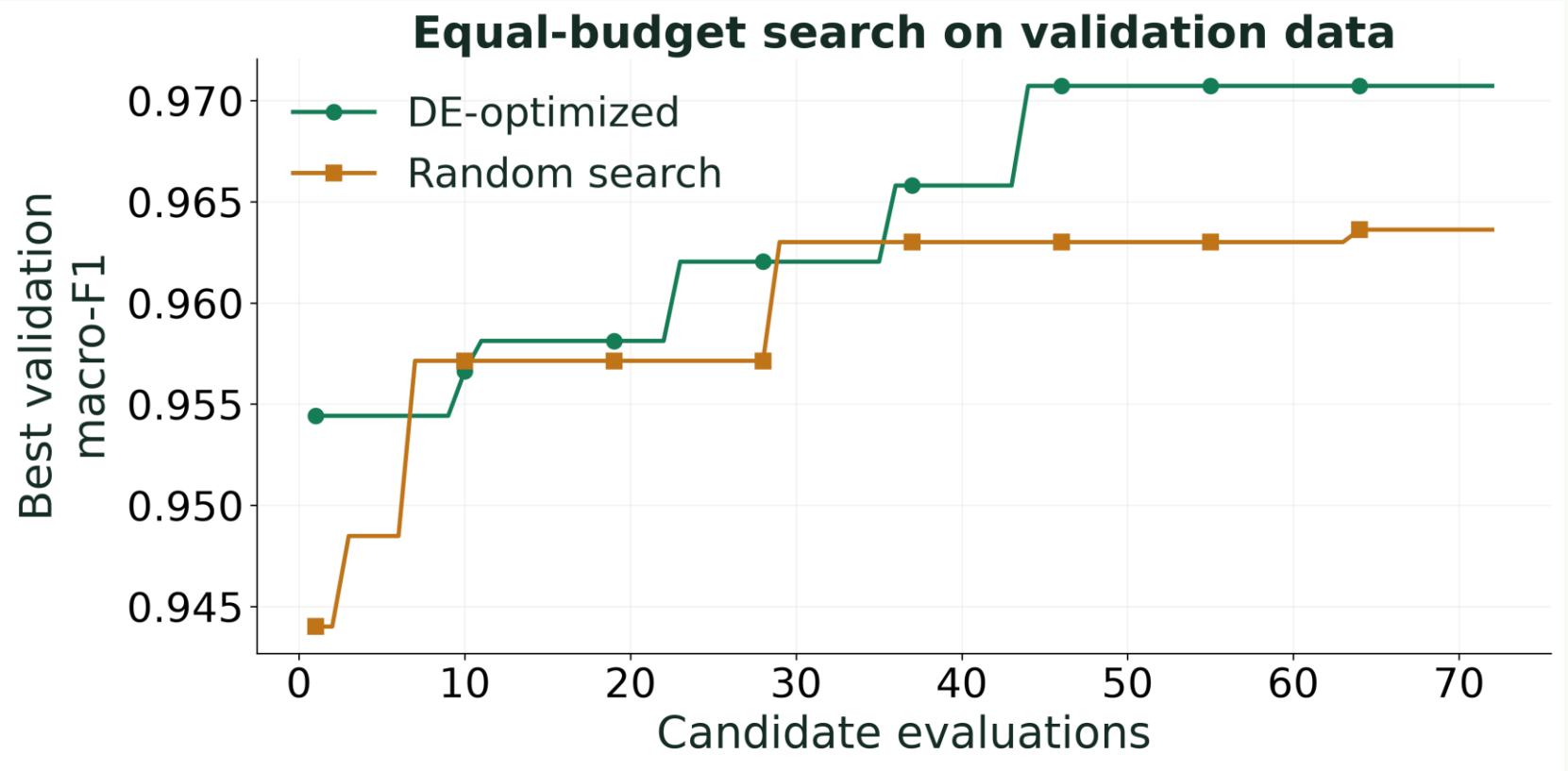
Implementation and Results (4/10)

SETTING	WHAT IT CONTROLS	RANGE SEARCHED
Learning rate	How fast the model learns	0.00003 to 0.01
Weight decay	Keeps the model simple to avoid overfitting	0.000001 to 0.01
Dropout	Randomly switches off neurons during training	0 to 0.6
Head width	Size of the final classification layer	0, 128, 256 or 512
Unfrozen blocks	How much of the pretrained network is retrained	0 to 9 blocks
Label smoothing	Stops the model from being over-confident	0 to 0.2
Augmentation	Random image changes that add variety	Level 0 to 15

Population 8 · 8 generations · 72 training runs for DE, and the same 72 for random search

DE VERSUS RANDOM SEARCH

Implementation and Results (5/10)



During the search, DE found the best validation score: 97.07% vs 96.36% for random search.

BEFORE AND AFTER DE

Implementation and Results (6/10)

BASELINE ACCURACY

97.43%

± 0.45 pp · 3 runs

DE ACCURACY

99.85%

± 0.06 pp · 3 runs

BASELINE MACRO-F1

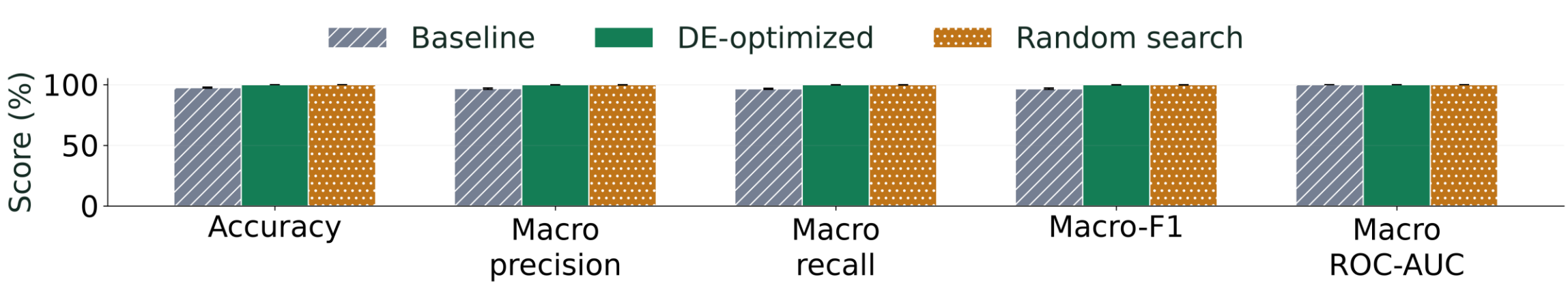
96.60%

± 0.54 pp · 3 runs

DE MACRO-F1

99.80%

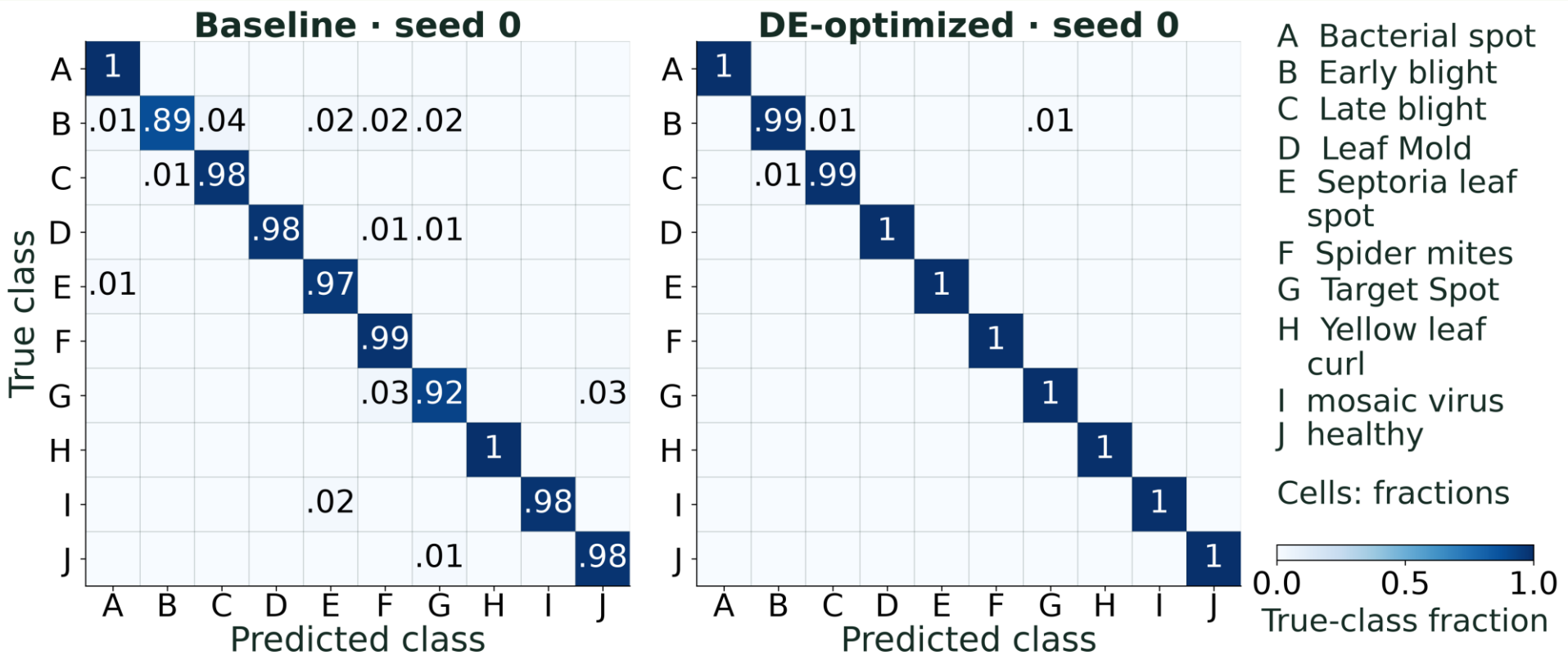
± 0.09 pp · 3 runs



DE cuts the baseline’s errors by 94.29% · both search methods reach 99.85% · macro-F1 = average score across all 10 classes

WHERE THE MISTAKES GO

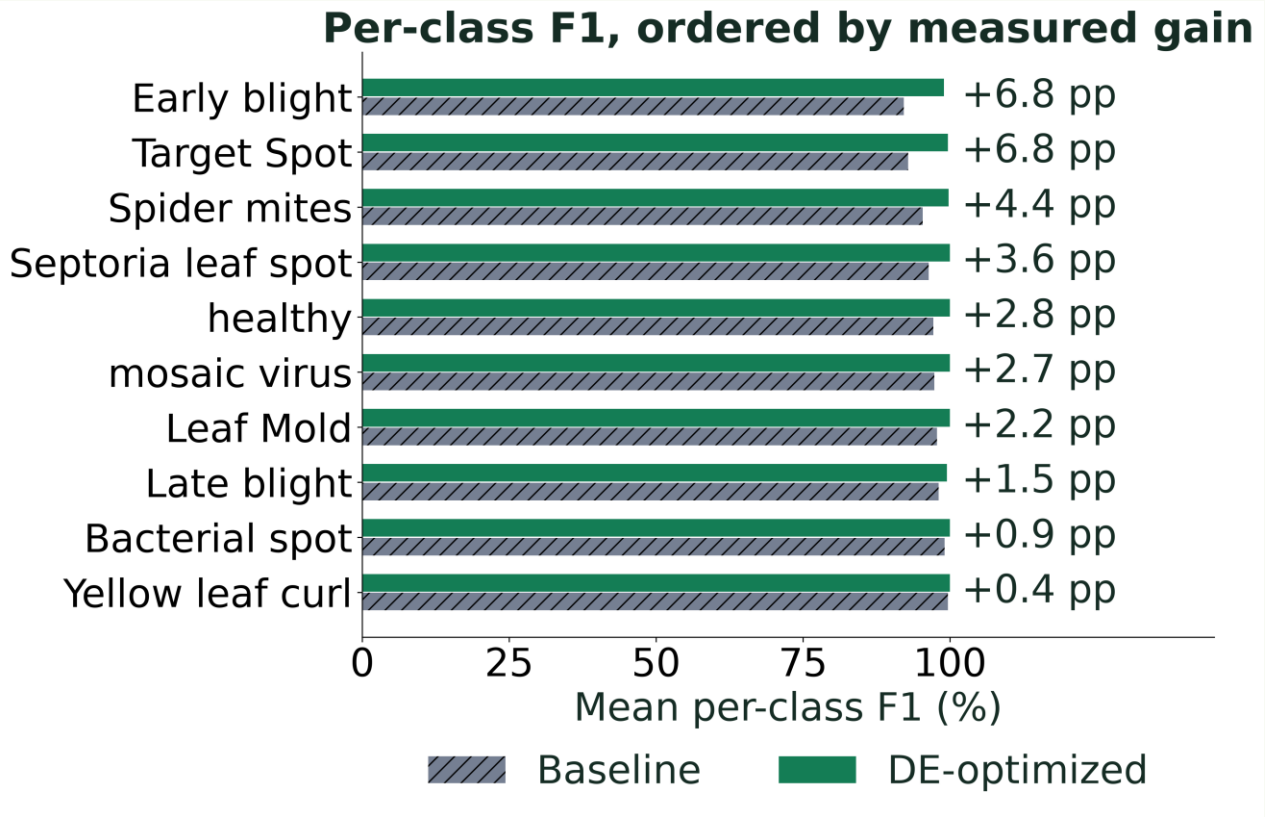
Implementation and Results (7/10)



Each row is the true disease. A strong diagonal means correct answers; DE removes most baseline mix-ups.

EVERY DISEASE IMPROVES

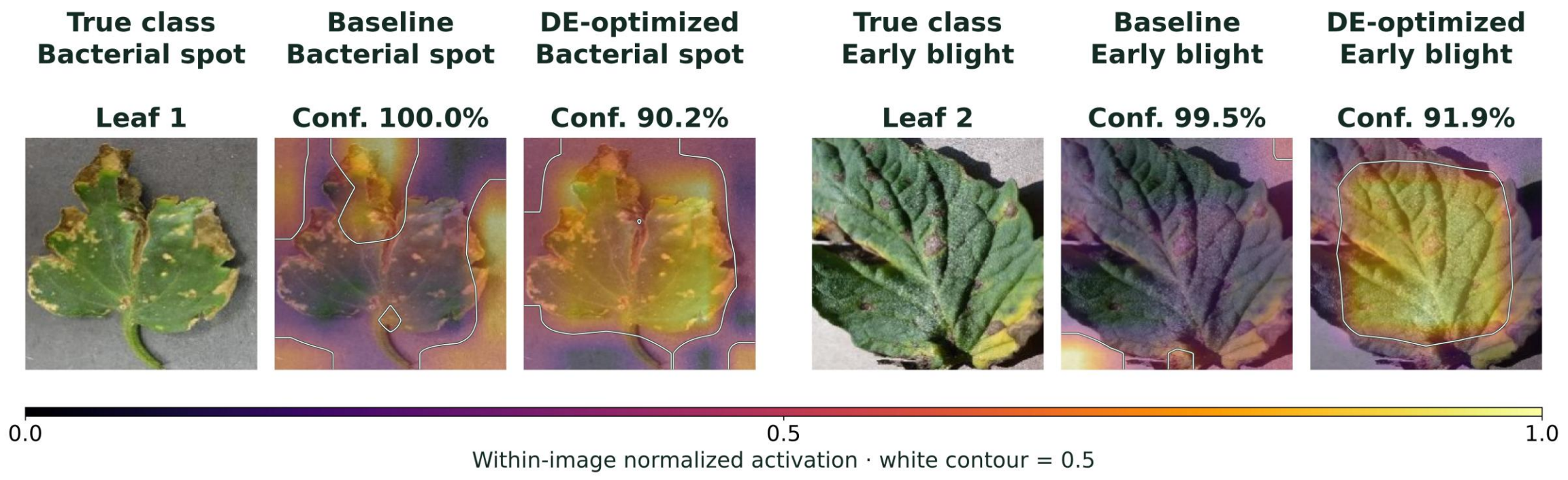
Implementation and Results (8/10)



All 10 classes improve with DE · largest gain: Early blight (+6.8 points)

WHERE THE MODELS LOOK

Implementation and Results (9/10)



Grad-CAM: a heatmap of where the model looked. Warm colours = more attention.

WHAT DE DISCOVERED

Implementation and Results (10/10)

SETTING	BASELINE	DE CHOICE	IN PLAIN WORDS
Unfrozen blocks	0	9	Retrain the whole network, not only the top
Learning rate	0.001	0.0013	Slightly faster learning
Label smoothing	0	0.11	Less over-confident training
Weight decay	0	0.00042	Mild simplicity penalty
Dropout	0.20	0.22	About the same
Head width	256	256	Unchanged

Key insight: DE learned that retraining all 9 blocks of the pretrained network pays off.

PLANTWISE · INTELLIGENCE, NATURALLY

Live Demo

An offline web app that runs on a normal laptop.

01

Diagnose

Pick a tomato leaf and compare the baseline and DE models side by side.

02

Explain

See the top-3 predictions and the Grad-CAM heatmap.

03

Evolve

Watch a DE population search a landscape, step by step.

HOW WE MEASURE SUCCESS

Performance Metrics and Benchmarking (1/5)

Three simple ways to score a disease classifier.

01

Accuracy

Share of test images labelled correctly.

02

Macro-F1

Average score across all 10 classes, so rare diseases count as much as common ones.

03

ROC-AUC

How well the model ranks the right disease above the wrong ones (1.0 = perfect).

IS THE GAIN REAL?

Performance Metrics and Benchmarking (2/5)

MODEL (RUN 1)	ACCURACY · LIKELY RANGE	MACRO-F1 · LIKELY RANGE
Baseline	97.13% – 98.24%	96.15% – 97.70%
Random search	99.74% – 100.00%	99.69% – 100.00%
DE-optimized	99.63% – 99.96%	99.48% – 99.95%

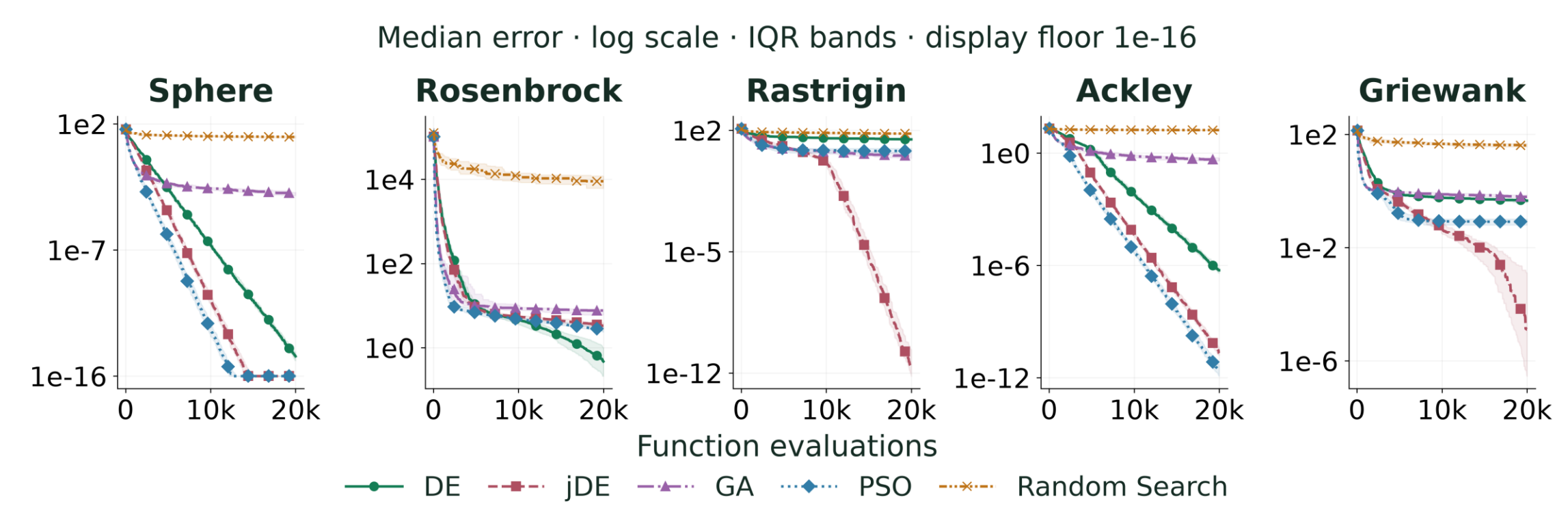
A statistical test (McNemar) confirms the gain is not due to chance ($p < 0.001$).

On the same test images, DE fixes 60 baseline mistakes; the baseline is right on only 3 that DE misses.

Likely range = 95% confidence interval from 1,000 resamples of the test set.

TESTING DE ON ITS OWN

Performance Metrics and Benchmarking (3/5)



5 standard test functions · 30 runs each · 20,000 tries per run · lower error is better

BENCHMARK RESULTS

Performance Metrics and Benchmarking (4/5)

FUNCTION	DE	jDE	GA	PSO	Random Search
Sphere	2.59e-15	1.27e-23	1.06e-03	2.59e-26	1.17e+01
Rosenbrock	4.72e-01	3.33e+00	7.66e+00	2.66e+00	8.99e+03
Rastrigin	3.00e+01	2.05e-12	3.30e+00	6.47e+00	6.44e+01
Ackley	5.46e-07	2.02e-11	4.47e-01	2.33e-12	1.66e+01
Griewank	4.57e-01	1.29e-05	6.22e-01	8.24e-02	4.10e+01

Median final error · lower is better · DE beats random search on 5 of 5 functions and is best on Rosenbrock

COST OF A BETTER RECIPE

Performance Metrics and Benchmarking (5/5)

MODEL	TRAINING (3 RUNS)	TIME PER IMAGE	PARAMETERS
Baseline	30.5 min	4.71 ms	4.34 M
Random search	66.1 min	4.80 ms	4.67 M
DE-optimized	84.5 min	4.74 ms	4.34 M

Search time: DE 45.9 min · random search 39.4 min

All three models predict at about the same speed and size.

WHAT WE ACHIEVED

Advantages and Limitations

01

Advantages

Accuracy 97.43% → 99.85%

DE written from scratch and unit-tested

Fair comparison with an equal budget

Works offline on a normal laptop

02

Limitations

Lab photos of single leaves

One crop: tomato

Search needs extra training time

NEXT STEPS

Future Scope

Build on these results with harder data and richer goals.

01

Field photos

Test on real farm photos taken with phones.

02

Multi-objective DE

Also reward robustness to blur and noise, reliable confidence and speed.

03

Mobile and adaptive

On-device app; self-tuning DE variants such as SHADE.

KEY SOURCES

References (1/2)

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KEY SOURCES

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Thank You / Questions

A healthier future.

One leaf at a time.

Plantwise · Intelligence, naturally.

github.com/Suraj-B12/BIO-INSPIRED-ALGO---AAT